2 Asset & Location

2.1 Overview

Asset is an item of economic value that a company own, benefit from, or has use of, in generating income. In EMS, an item is treated as asset when user wants to track its moving history, and maintenance work can occurs toward it.

Asset can be either rotating or non-rotating. Rotating assets are assets that are interchangeable, such as motors, pumps, or fire extinguishers. In EMS, remarkable difference between rotating asset and non-rotating asset is that rotating assets have both an asset number and an inventory item number, while non-rotating just have an asset number, without an inventory item number. The item number lets user to track assets as a group as they are moved in and out of inventory and other types of locations, while the asset number is useful to track individual instances of asset as it is moved from one location to another and from one site to another.

Here is example to help understanding inventory item number and asset. A company might have four identical (same make, same model) centrifugal pumps, so all four pumps would have the same item number. To track the use, repair, and locations of each individual pump, each pump would have its own, unique asset number.

Location is generally defined as a place where assets are operated, stored, and repaired. Generally locations are defined as a means of tracking assets. Another consideration, a maintenance work can be done toward **Locations**.

2.2 Classification Application

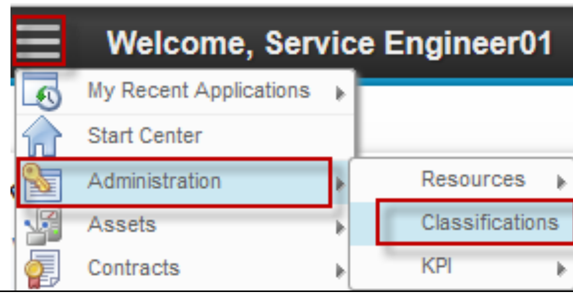
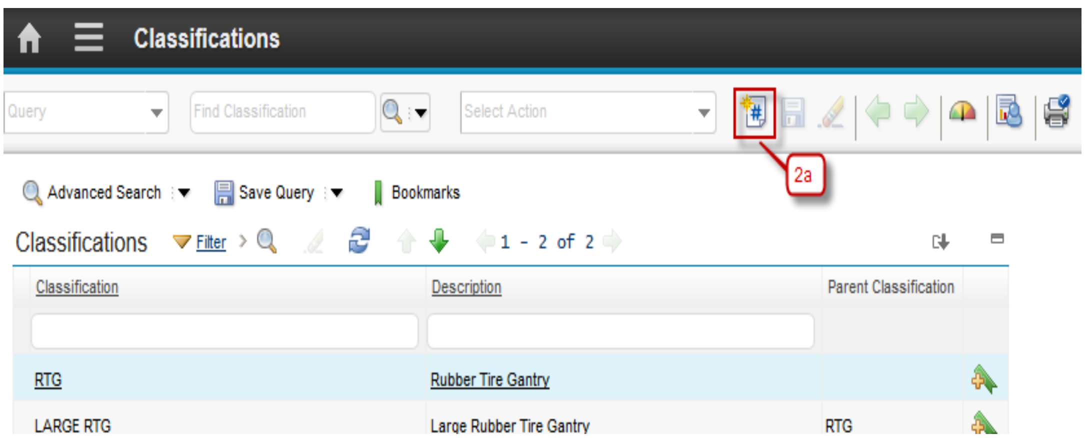
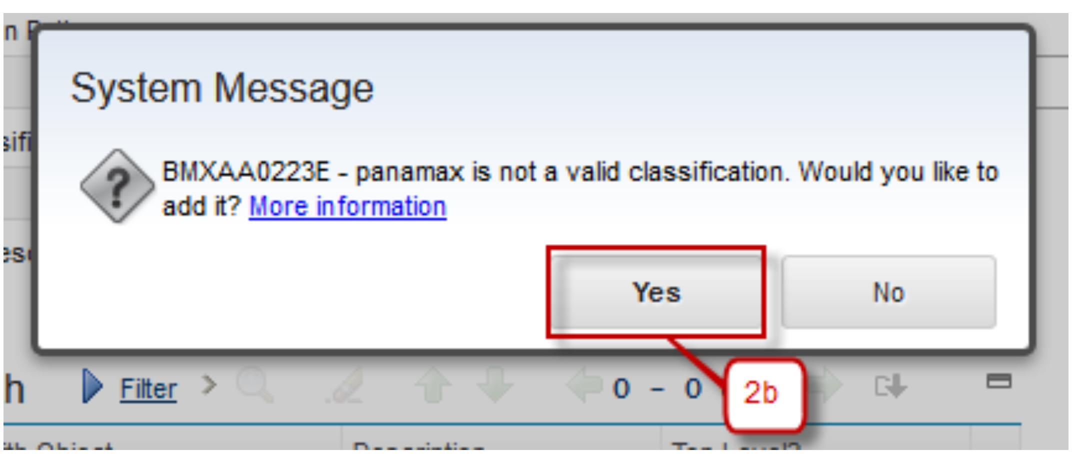
The Specifications tab on asset and location records can be used to add Classification information to asset and location records. Specification is available on item data too. Using Classifications lets user to structure assets and locations in organized hierarchies so that:

- Records can be easily located via Classifications.
- Records are not duplicated unintentionally.
- Asset and item lists are consistent with vendor's lists.

Classifications use attributes to define the classification category. When you list a rotating item that has a classification on an asset or location record, Maximo copies the Classification and Attributes to **Specifications** tab of the asset or location record.

2.2.1 Create Classifications

Classification creation can be performed by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to create Classification:

Step	Action
1. Go to Classification application by click (GoTo) – **Administration – Classifications.**	
2a. Click **New Classification.**   2b. If the classification that inputted above doesn't valid, the message box to added the classification will appear. Click button **Yes** and then classification will be added.	

3a. Input the Classification name.

3b. Input the Classification description.

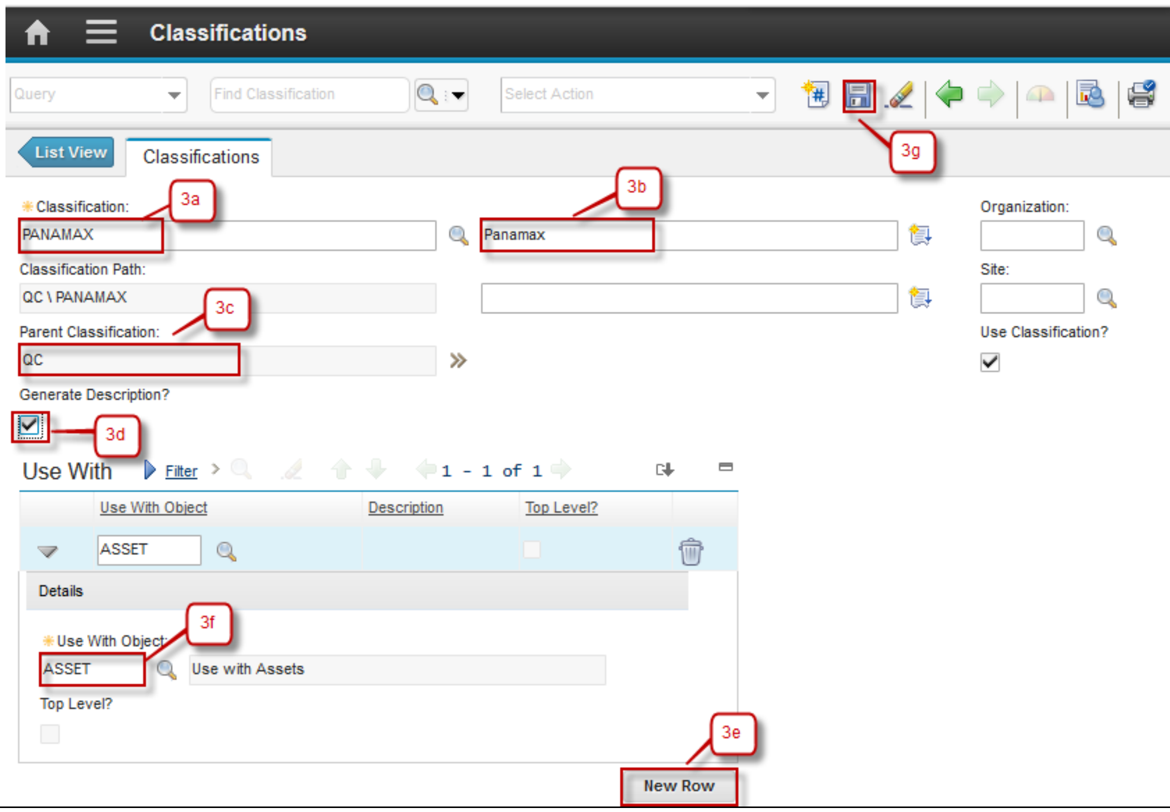
3c. Input the Parent Classificication if this classification has a parent.

3d. Default from **Generate Description** is checked, if description classification don't want generated when the classification in use make sure the check box is unchecked.

3e. Click **New Row** to add the object that will use this classification.

3f. Input the object that will use this classification.

3g. Click **Save.**



2.2.2 Define Attributes to a Classification

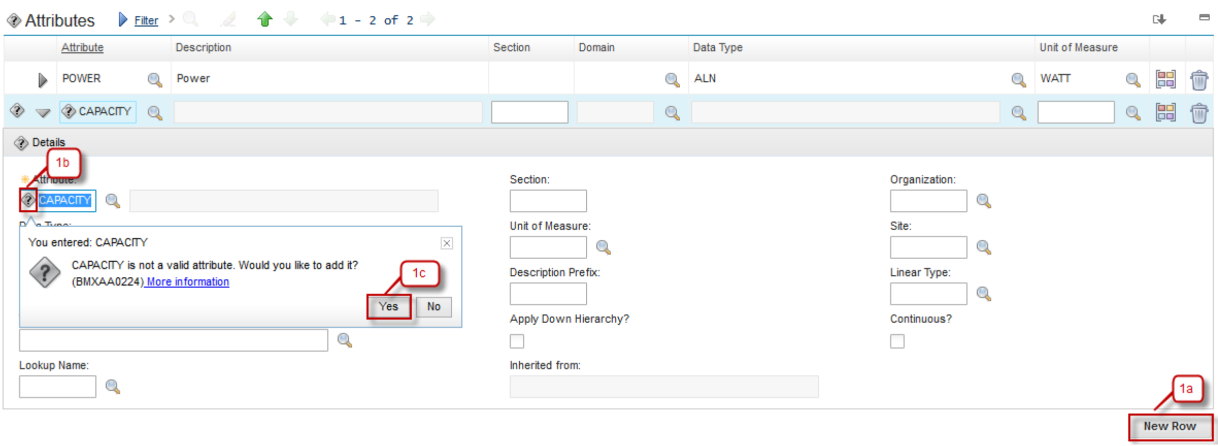
Defining attributes to a classification can be performed by *Service Engineer*, *CHE Engineer*, and *Engineering Manager*. Below are step by step to Define Attribute to a Classification:

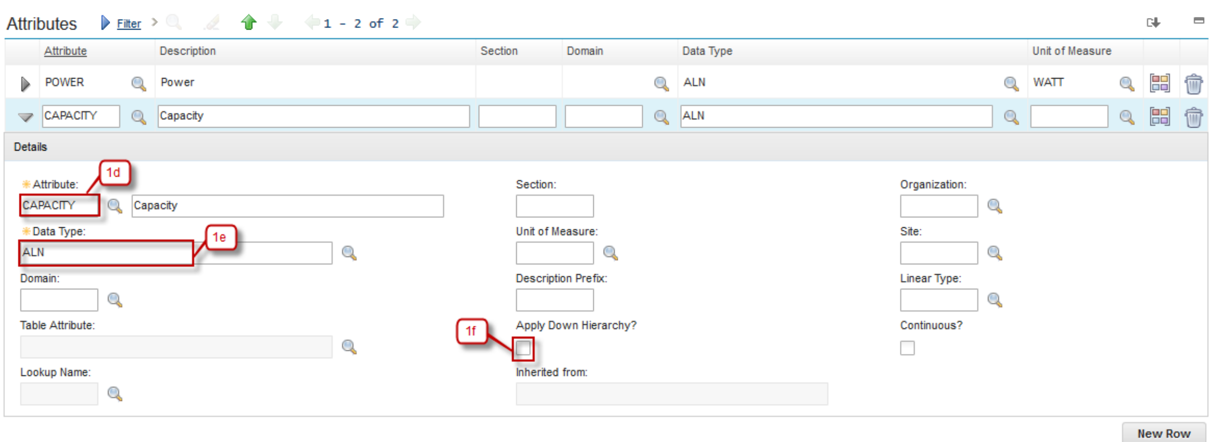
Step	Action
1a. Click **New Row** to add attribute on this classification.	
1b. If the attribute you added is a new attribute, question mark icon will appear.	
1c. Click **Yes** to add the attribute.	

1d. Input the Description of that attribute.

1e. Input the Data Type of that attribute.

1f. Check the **Apply Down Hierarchy** checkbox if you want this attribute will appear in any child classification of this classification and click **Save**.





2.3 Locations Application

Locations application is used to add, view, modify, and delete location records for assets, and organize these location into systems. Systems are used to maintain location's hierarchical. When location is organized into systems, user can quickly find a location using the drilldown and then identify the assets at a specific location.

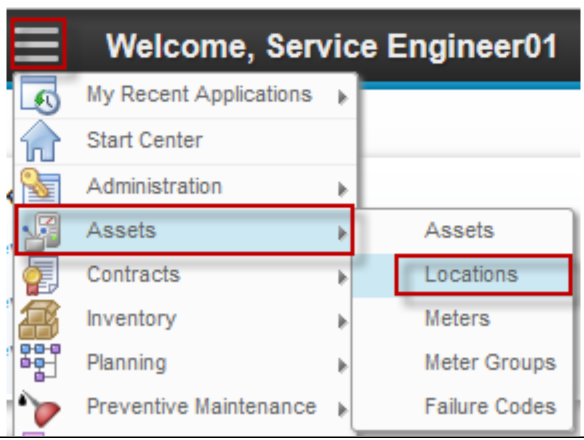
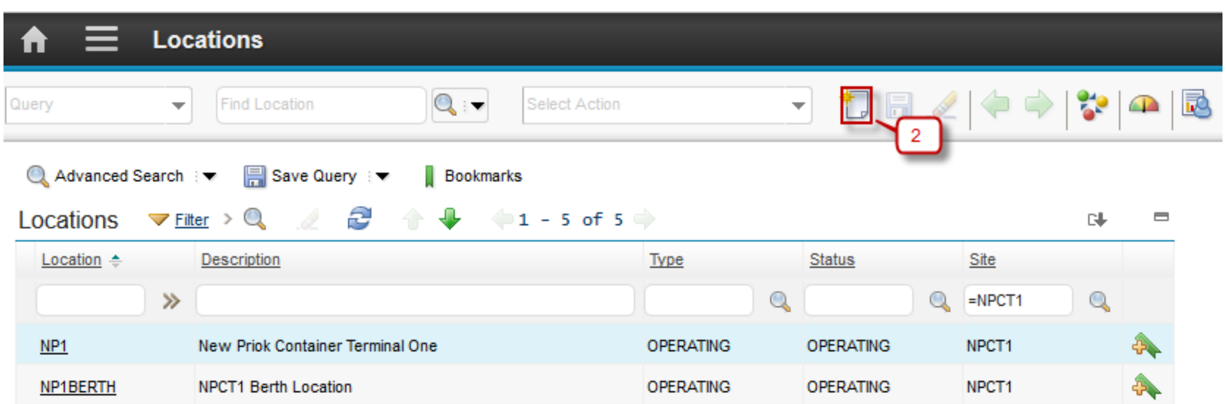
Locations are categorized into several types:

- Courier - This is an inventory location, used to indicate an individual who is responsible for inventory items while in transit from one inventory location to another.

- Holding - This is an inventory location, used to hold items received with a status of WINSP (waiting for inspection). One site can only have one holding location.
- Operating - This is an asset location, used to indicate a location where assets are located or equipment is operated.
- Repair - This is an asset location, used to indicate a location where assets are repaired or refurbished.
- Salvage - This is an asset location, used to indicate a location used to store retired or decommissioned assets.
- Vendor - This is an inventory location, used to indicate the source of items, material, and rotating assets, and may be used to indicate the location of an asset returned to the vendor for repair or refurbishment.

2.3.1 Create Locations

Location creation can be performed by *Service Engineer*. Below are step by step to Create Locations:

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1. Go to Locations application by click  (GoTo) – Assets – Locations.</td><td>
</td></tr>
<tr><td colspan="2">2. Click  New Locations .

</td></tr>
</table>

3a. Input the Locations name.

3b. Input the Locations description.

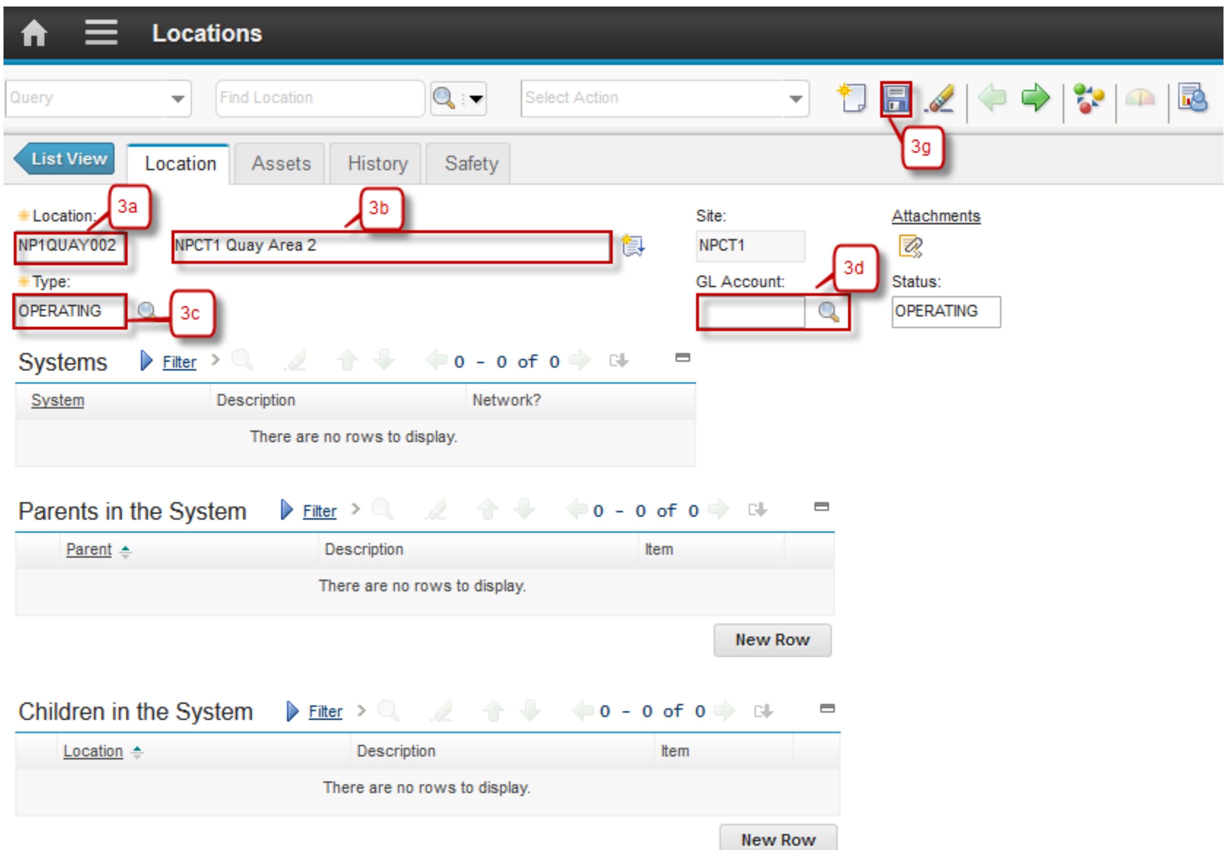
3c. Select the type of Locations, default type is ‘OPERATING’.

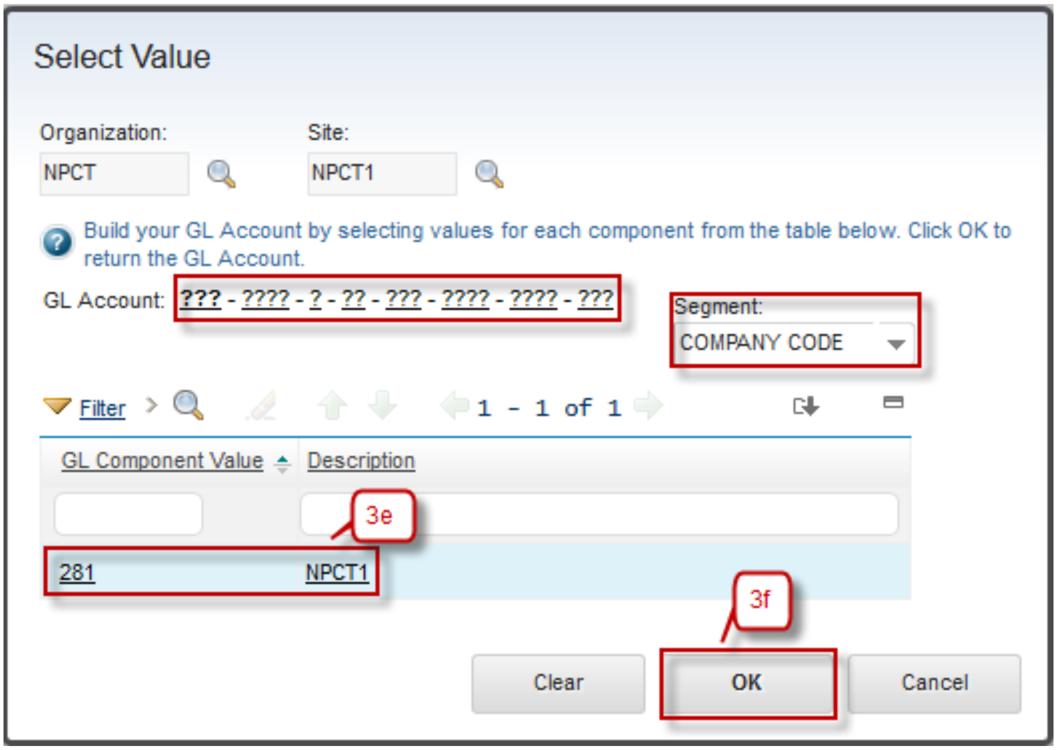
3d. Define GL Account of Locations, click button

3e. Select account value for every segment of GL Account.

3f. Click **OK**.

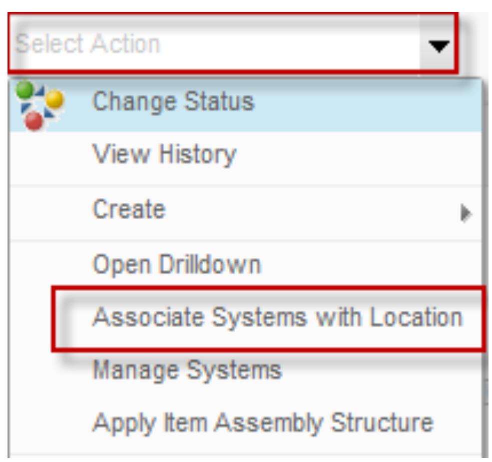
3g. Click **Save.**





2.3.2 Associate Systems with Locations

Below are step by step to Associate Systems with Locations by *Service Engineer*:

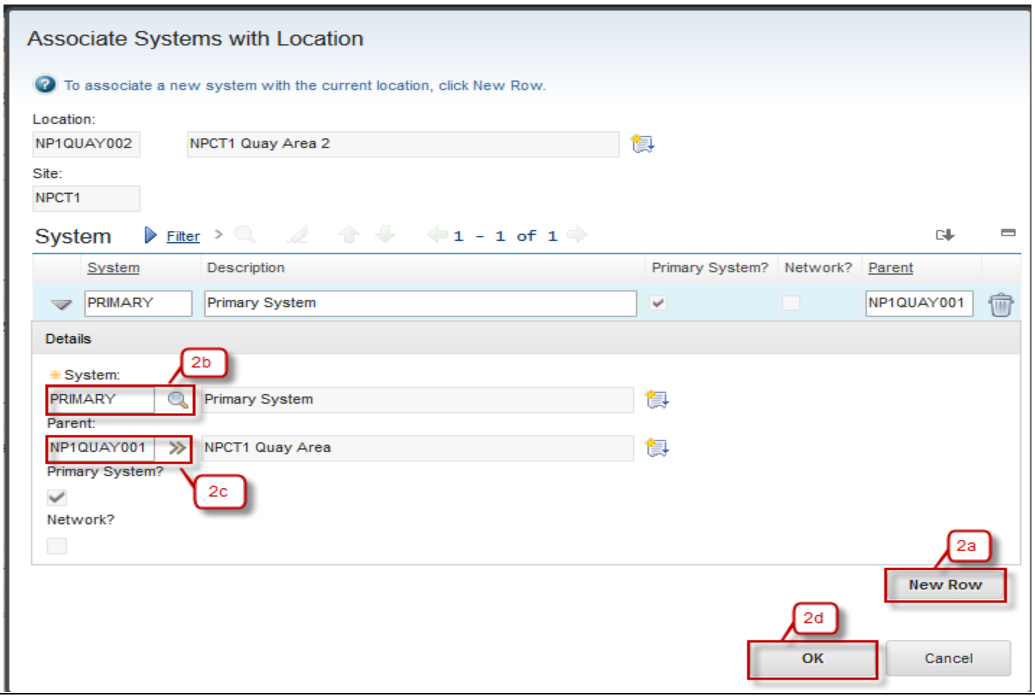
Step	Action
1. Go to **Select Action – Associate Systems with Location.**	

2a. Click **New Row.**

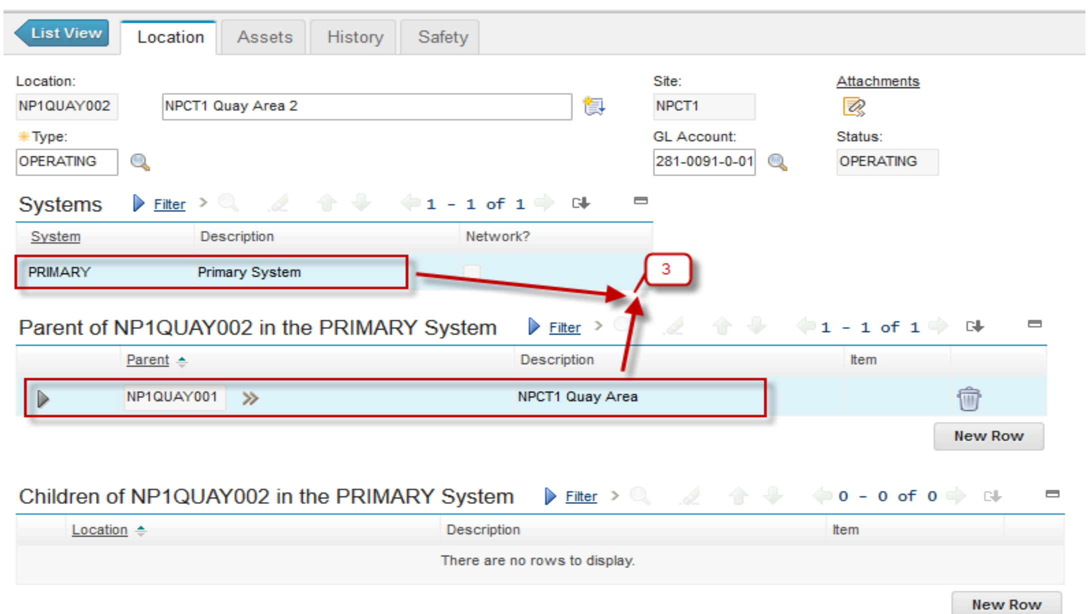
2b. Select System of Locations.

2c. Select Parent of Locations.

2d. Click button **OK.**

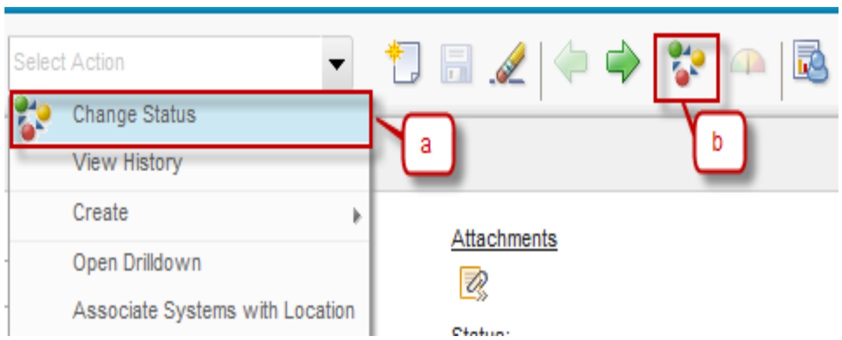
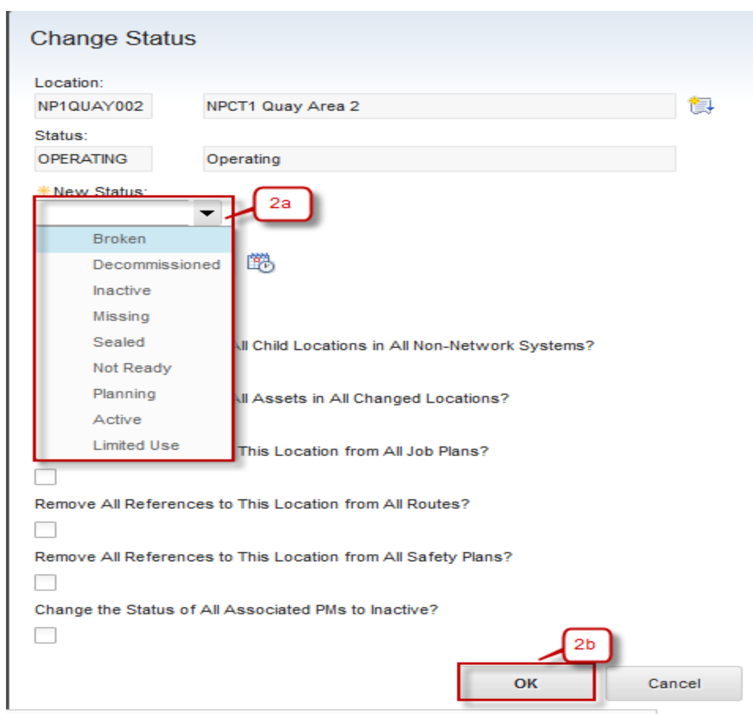


3. Field System and Parent will defined

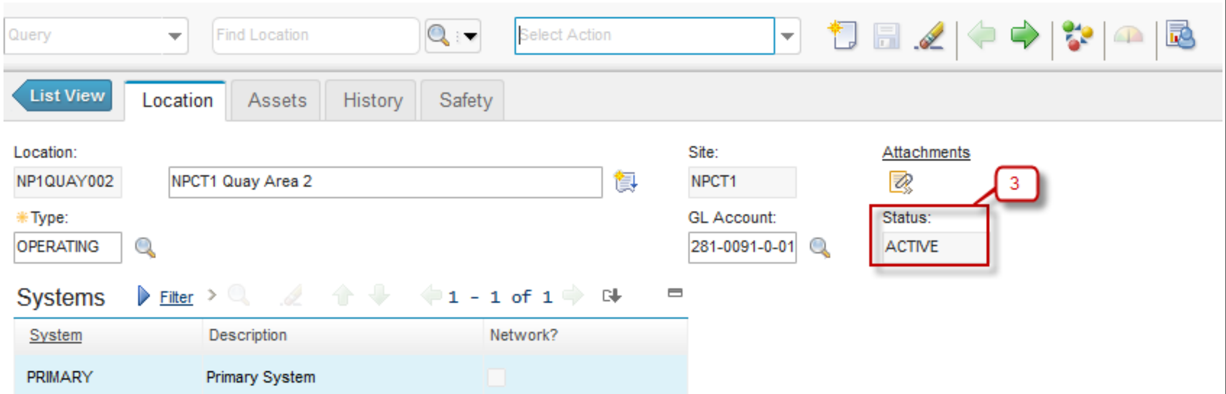


2.3.3 Change Status of a Location

Below are step by step to Change Status of a Location by *Service Engineer*:

Step	Action
1. Change Status of Person have two ways: a. From **Select Action → Change Status** b. Click button  	
2. Dialog box **Change Status** will be appeared, see the figure below : 2a. Select New Status of Locations. 2b. Click **OK.** 	

3. Status will be change to **ACTIVE**



2.4 Meters & Meter Groups Application

A meter record in EMS is virtual identifier used to record measurements. meter records is associated with assets and locations to record measurements, for example run hours, temperature, or pressure, that are used to track asset and location performance. **Meters** application is used to add, view, modify, and delete meter records.

Meters can have one of three types:

- Continuous - Meters where the readings increase continuously, for example odometers. These meters might be used to track miles, hours, engine starts, pieces produced, fuel consumed, and other continuous readings.
- Gauge - Meters where the readings may fluctuate, for example thermometers or pressure gauges. These meters might be used to track temperature, pressure, noise levels, oil levels, and other readings that fluctuate,
- Characteristic - Meters where an observed state is being tracked, for example a color change. These meters might be used to track Brick/Refractory Color (yellow, orange, white), or oil color (clear, turbid, dark).

Meter readings may be recorded for a variety of reasons, including:

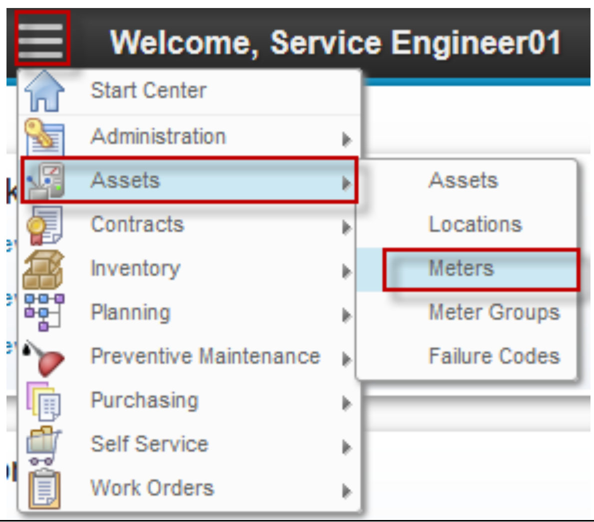
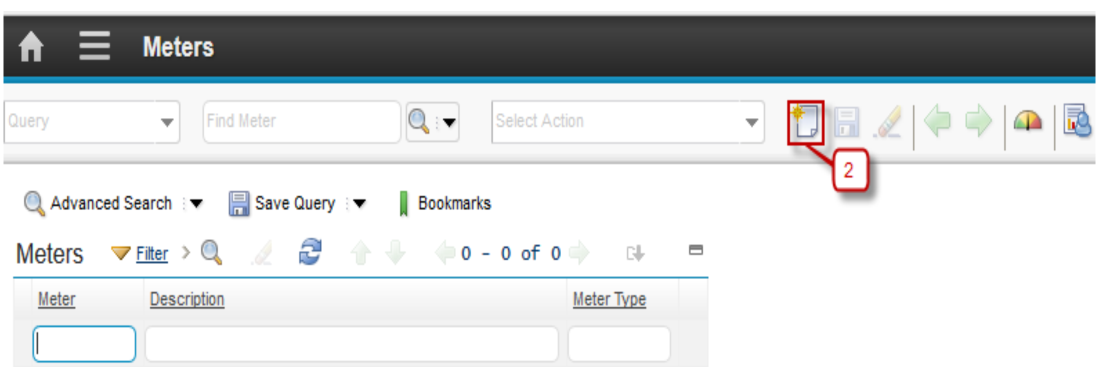
- determining the next time an asset is due for maintenance
- performing efficiency comparisons between similar asset performing similar work
- tracking production costs, for example tons per hour in the mining industry
- tracking service costs, for example odometer readings for a trucking fleet
- tracking efficiency indicators, for example miles per gallon
- conforming to regulatory requirements, for example maintaining cradle to grave details of time controlled components in the aviation industry

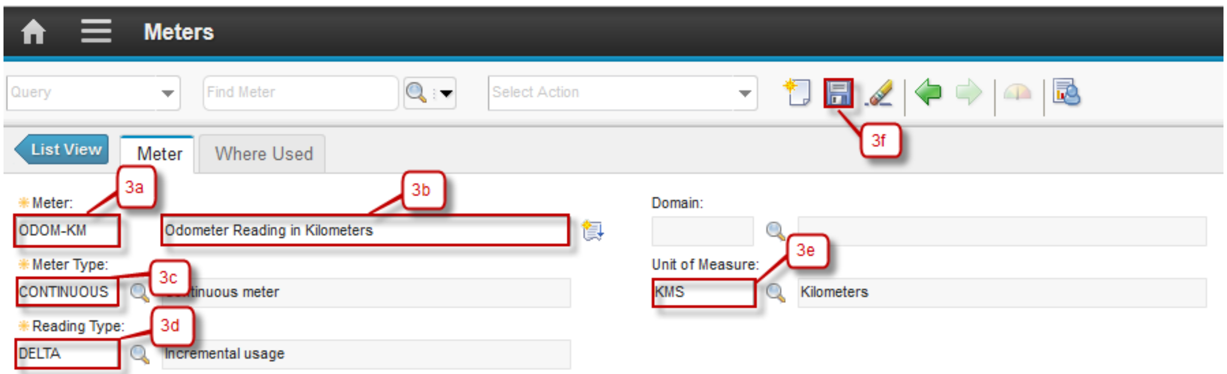
In **Meters** application, when user create a meter record, user can define the type of meter, the unit of measure used by the meter, and the type of reading (delta or actual) that the meter should record. Meter records can be associated with assets, inventory items, and locations, and can be used to drive condition monitoring and preventive maintenance. Meters can be applied individually, or as part of a meter group.

A meter group is two or more meters that are commonly used on the same types of assets or locations. **Meter Groups** application is used to create groups for meters that are commonly used together. For example all of the vehicles in a fleet use meters to track mileage, usage, fuel, and oil consumption. Rather than attach these four meters to a vehicle one at a time, user can define a meter group and use it to associate all four meters to the vehicle at the same time.

2.4.1 Create Meters

Below are step by step to Create Meters by *Service Engineer*:

Step	Action
1. Go to Meters application by click  **(GoTo) – Assets – Meters.**	
2. Click  **New Row.** 	

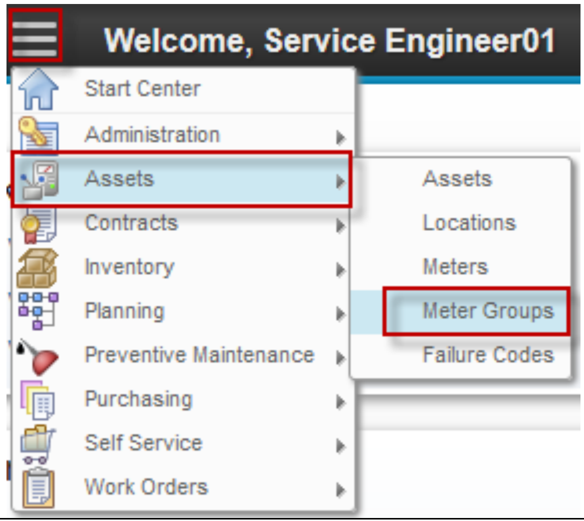
3a. Input the Meter name. 3b. Input the Meter description. 3c. Select the type of Meter. 3d. Input the Reading Type. • If the type of Meter is 'CONTINUOUS', field Reading Type is mandatory. • If the type of Meter is 'CHARACTERISTIC', field Domain is mandatory. 3e. Input the Unit of Measure 3f. Click **Save.** 

2.4.2 Create Meter Groups

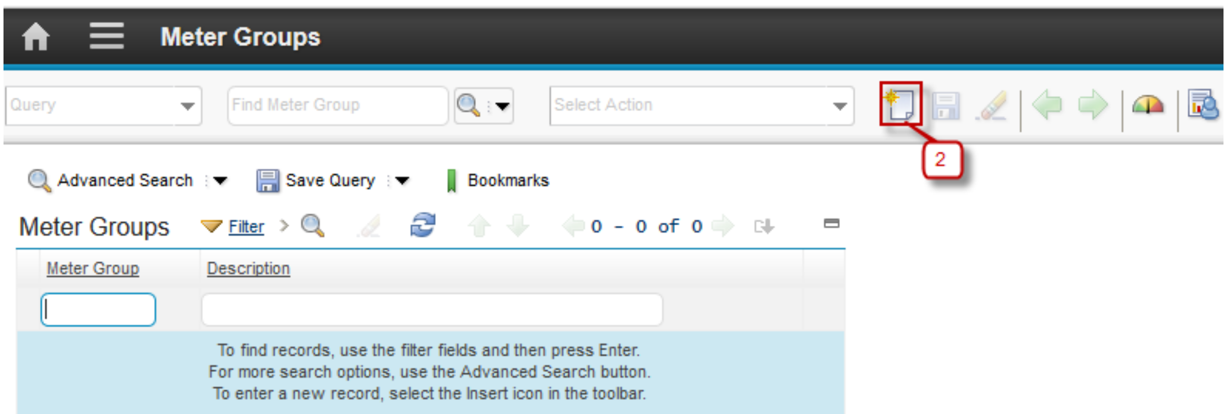
Below are step by step to Create Meter Groups by *Service Engineer*:

Step	Action

1. Go to Meters application by click (GoTo) – **Assets – Meter Groups.**



2. Click **New Row.**



3a. Input the Meter Group name.

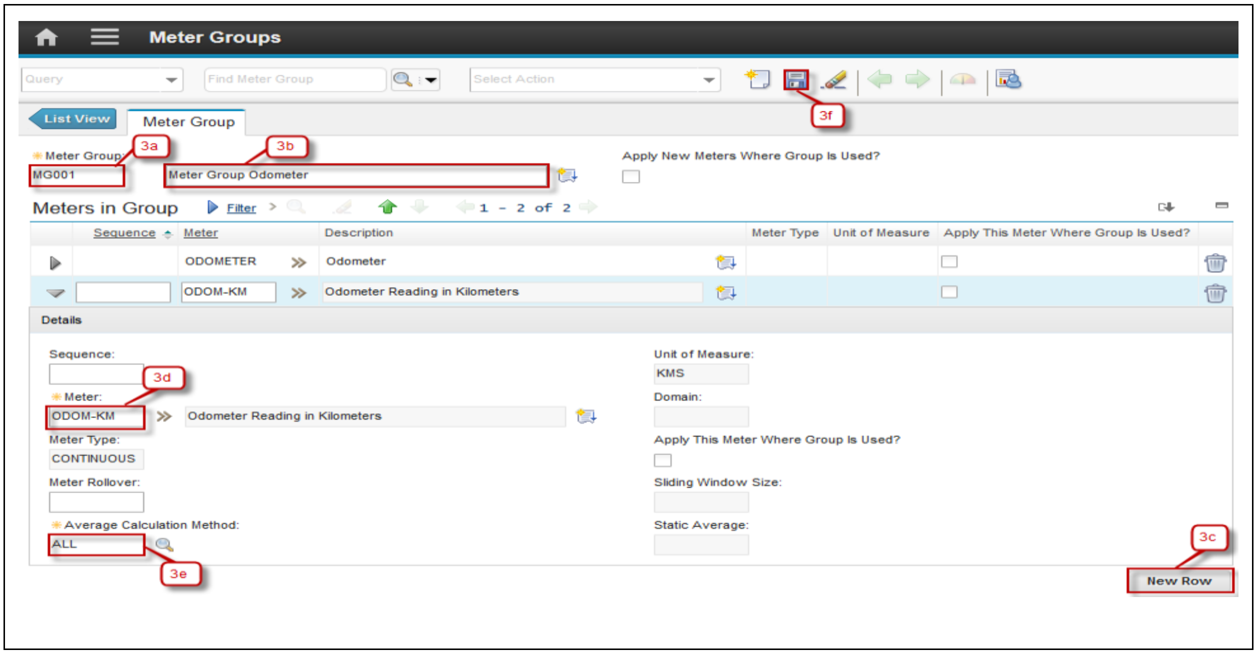
3b. Input the Meter Group description.

3c. Click **New Row**.

3d. Select Meter using button »

3e. Select Average Calculation Method using button »

3f. Click **Save.**



2.5 Assets Application

Asset application is used to create and store asset numbers and corresponding information, such as parent, location, vendor, up/down status and maintenance costs.

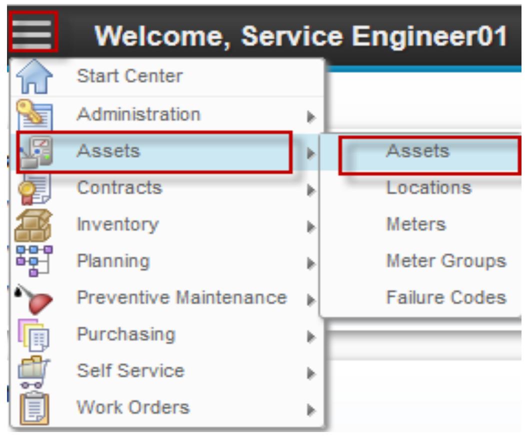
In **Asset** application user can build asset hierarchy, an arrangement of buildings, departments, assets and subassemblies. The asset hierarchy provides a convenient way to roll up maintenance costs so that user can check accumulated costs at any level, at any time. It also makes it easy to find a particular asset number.

2.5.1 Create Assets

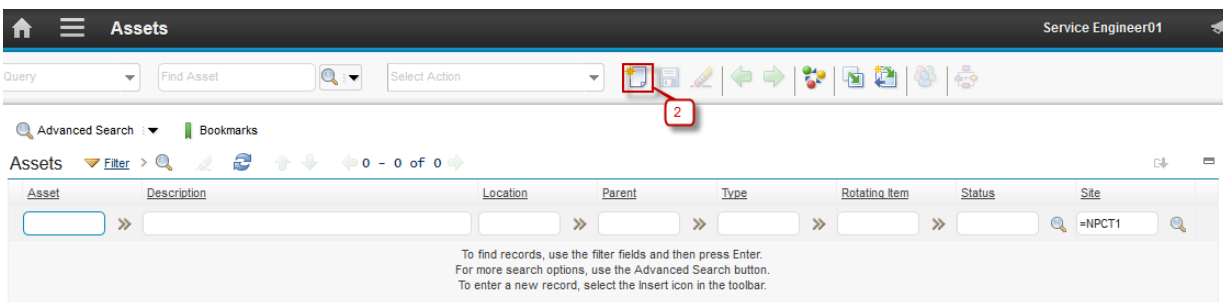
Asset creation can be done by *Service Engineer*. Below are step by step to Create Assets:

Step	Action

1. Go to Assets Application by click (GoTo) – **Assets – Assets.**



2. Click **New Row.**



3a. Input the Asset name and make sure default status is 'PLANNING'.

3b. Input the Asset description.

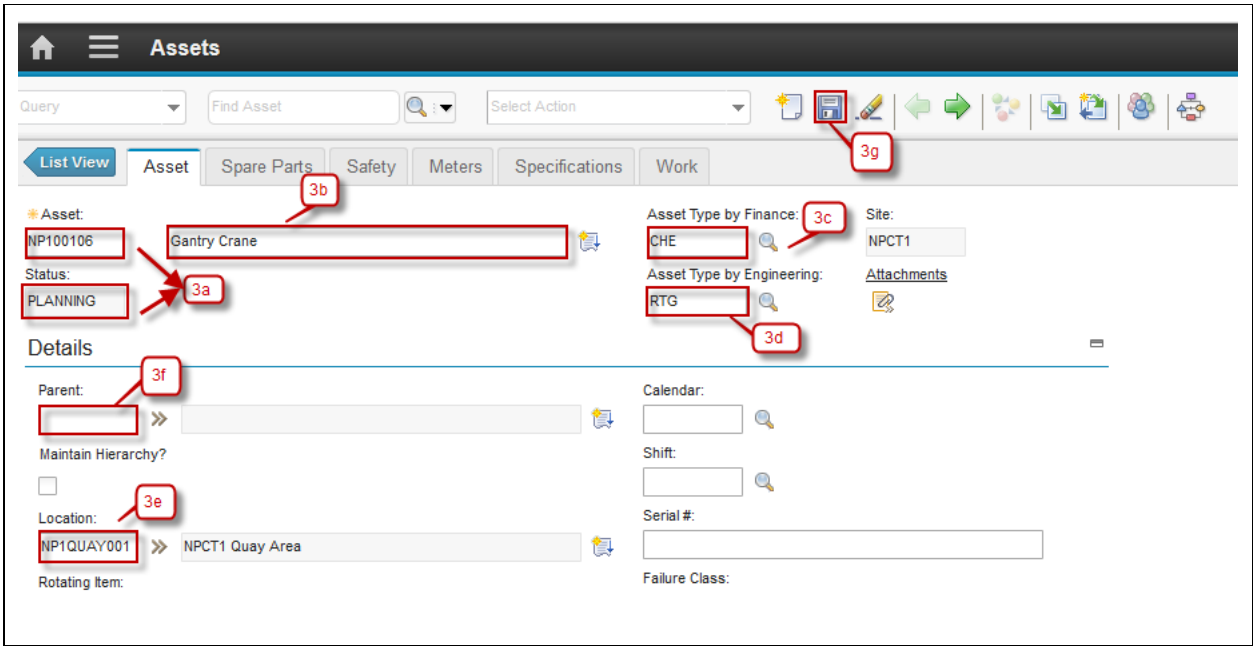
3c. Select Asset Type by Finance using button .

3d. Select Asset Type by Engineering using button .

3e. Input Parent of Asset if the Asset have a parent.

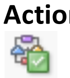
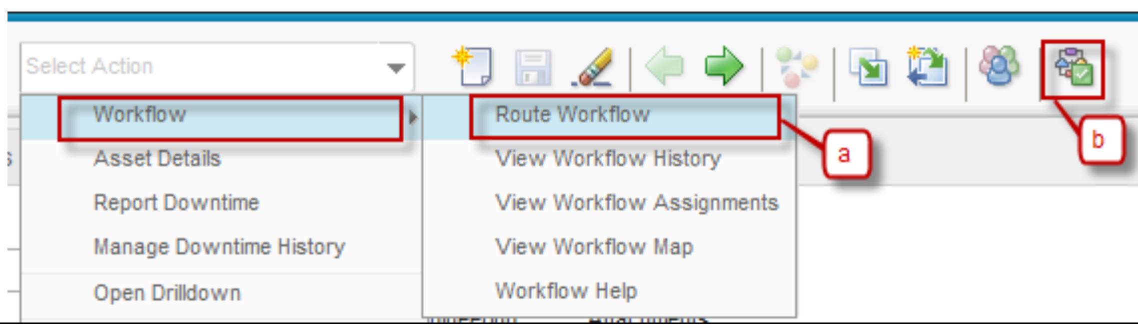
3f. Input Location of Asset**.**

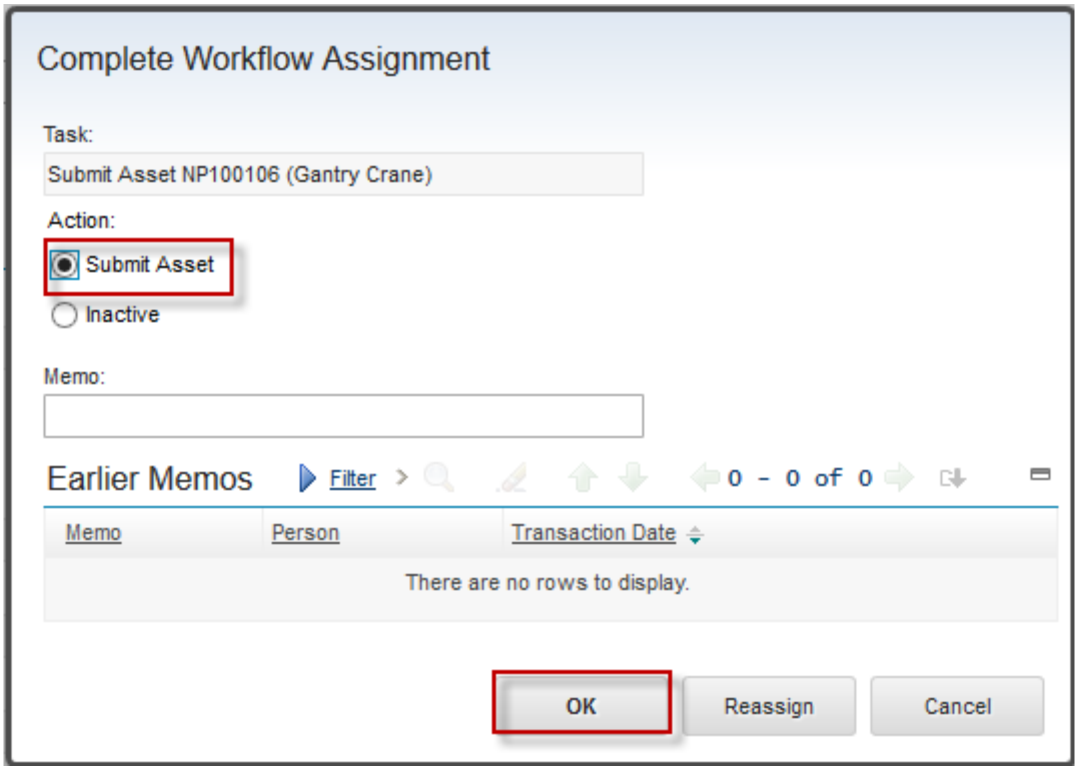
3g. Click **Save** and then after save the Asset button **Route Workflow** will be started automatically



2.5.2 Asset Creation Approval by CHE Engineer

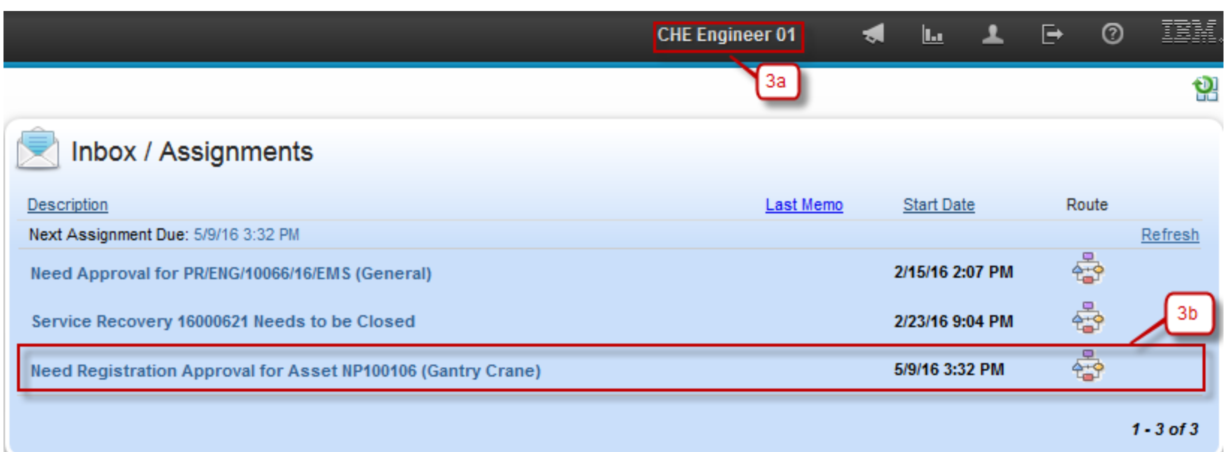
Created Asset must be approved by *CHE Engineer*. Below are step by step to Asset Creation Approval:

Step	Action
1. Asset Creation Approval have two ways: a. From **Select Action → Workflow → Route Workflow** b. Click button  	
2. Click **Route Workflow** and dialog box will be appeared, see the figure below:	



3a. Login as CHE Engineer 01.

3b. Assigment will be appeared in column **Inbox/Assignments** for approval Asset.

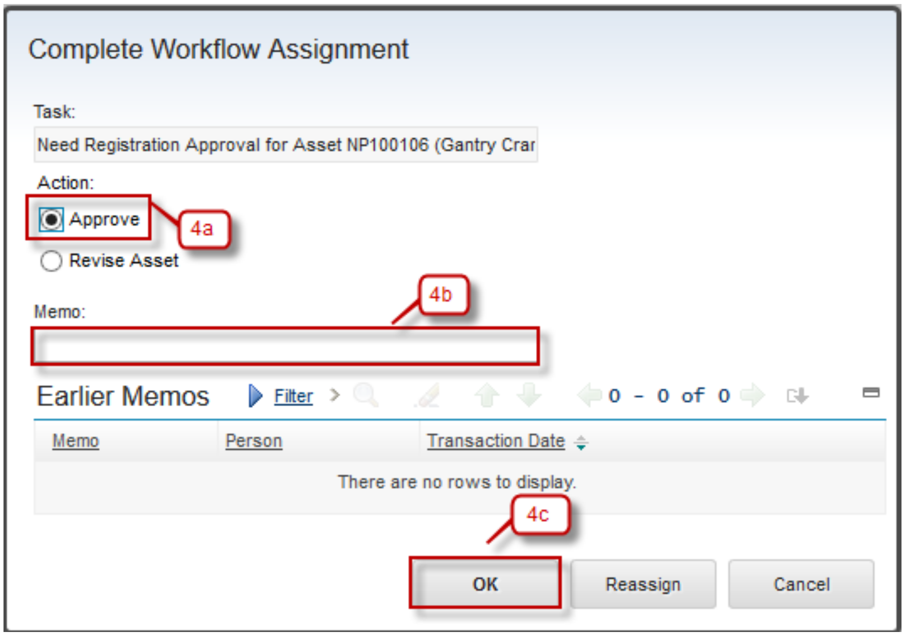


4. Click button **Route Workflow** and dialog box will be appeared see the figure below:

4a. Select Action Approve

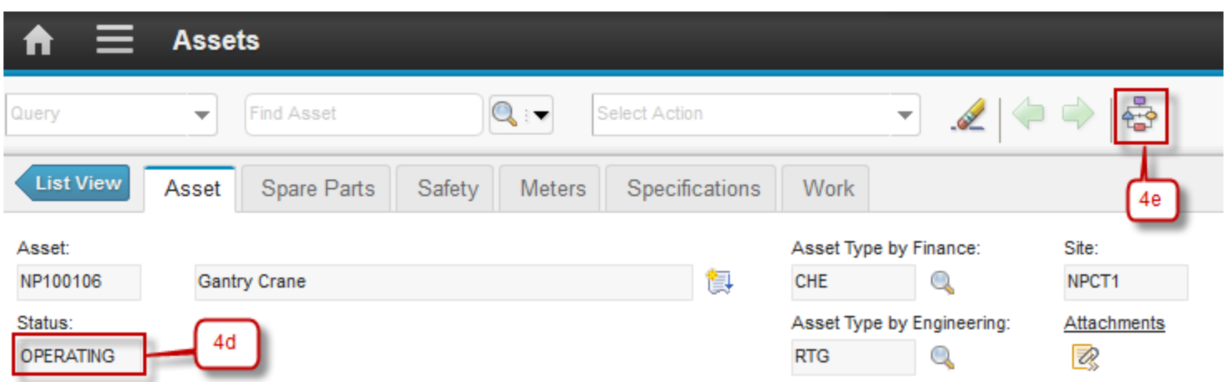
4b. Input Memo (if needed)

4c. Click **OK**



4d. Status changed to 'OPERATING'.

4e. Workflow stopped.

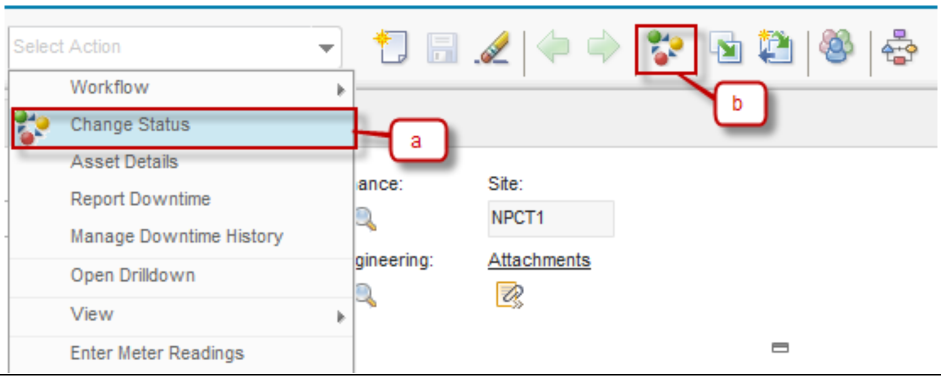


2.5.3 Change Status of an Asset

Sometimes, there is a needs to change the status of an asset. For example, when an asset is broken or decommissioned, user should change the status, to indicate that the asset is now in broken or decommissioned status. Below are step by step to Change Status of an Asset by *Service Engineer*:

Step	Action

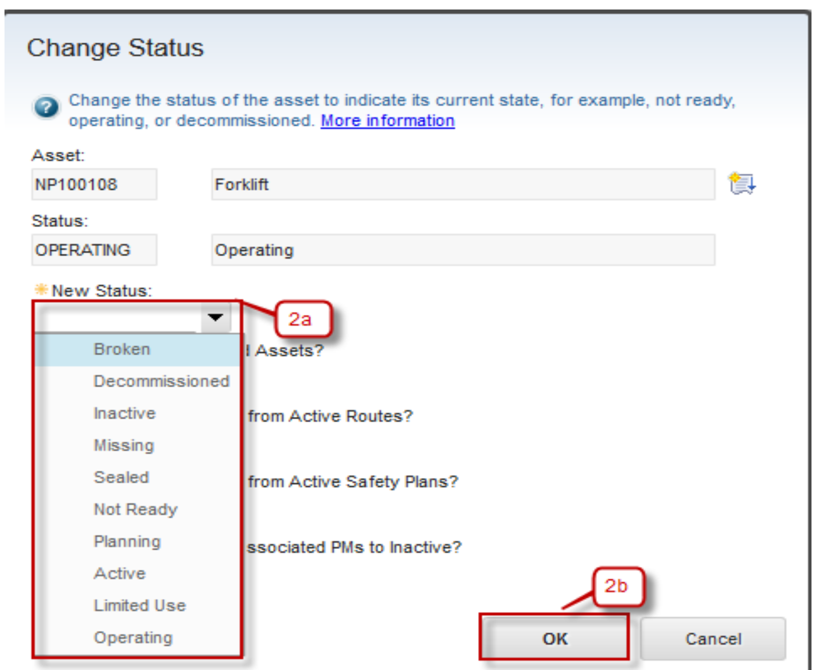
1. Change Status of Person have two ways:
    a. From **Select Action → Change Status**
    b. Click button 



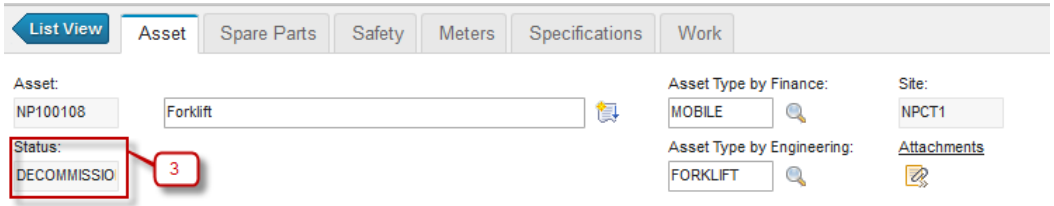
2. Dialog box **Change Status** will be appeared, see the figure below :

2a. Select New Status of Asset.

2b. Click **OK.**



3. Status changed to **DECOMMISSIONED**

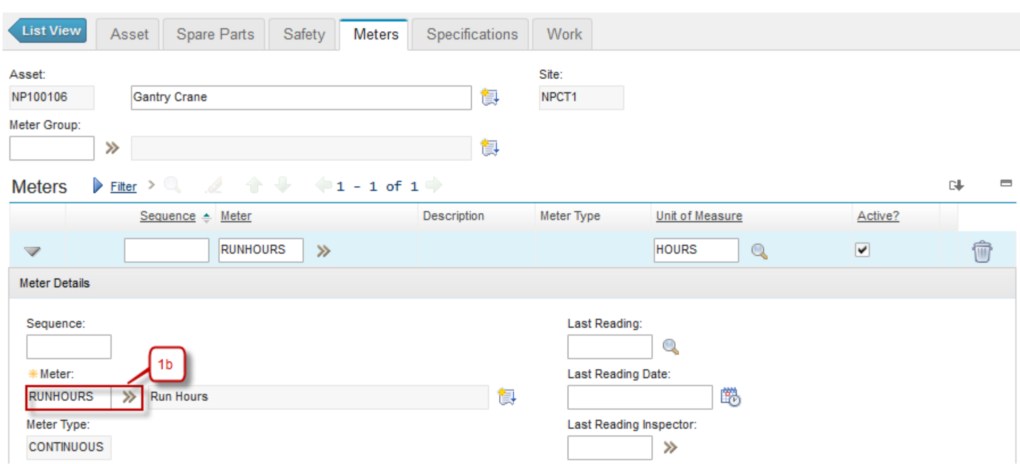


2.5.4 Assets Meters Registration & Meter Readings

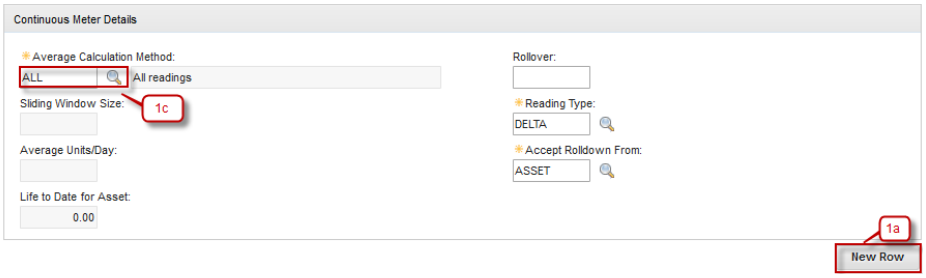
User can register all meters that apply to an asset in **Assets** application. Meters registration in an asset can be performed by registering a meter group that will automatically register all meters in meter group to the asset, or manually registering each meter to the asset.

In **Asset** application, user also can perform **Enter Meter Reading** action to report meter readings for the selected asset. This meter reading will be used as a measurement about the assets performance based the meters. This meter reading can be used to determine when a maintenance toward those asset should be performed. As additional notes, Enter Meter Reading action can be performed in **Work Order Tracking** application too (This application will be discussed later).

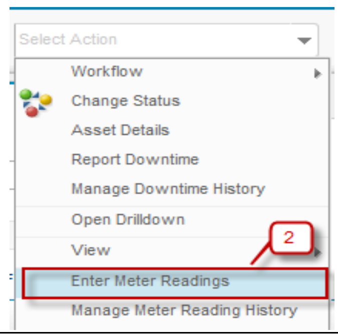
Below are step by step to register a meter to an asset, and performs **Enter Meter Reading** action, by *Service Engineer*:

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td colspan="2">
1. Select one of Asset, go to Tab Meters .
1a. Click New Row.
1b. Select the Meter to use in Asset using button », field Meter Type and field Unit of Measure automatically fill.
1c. If the type of Asset is ‘CONTINUOUS’ , field Average Calculation Method is mandatory.


</td></tr>
</table>



2. To perform enter meter reading action, choose **Select Action → Enter Meter Readings.**

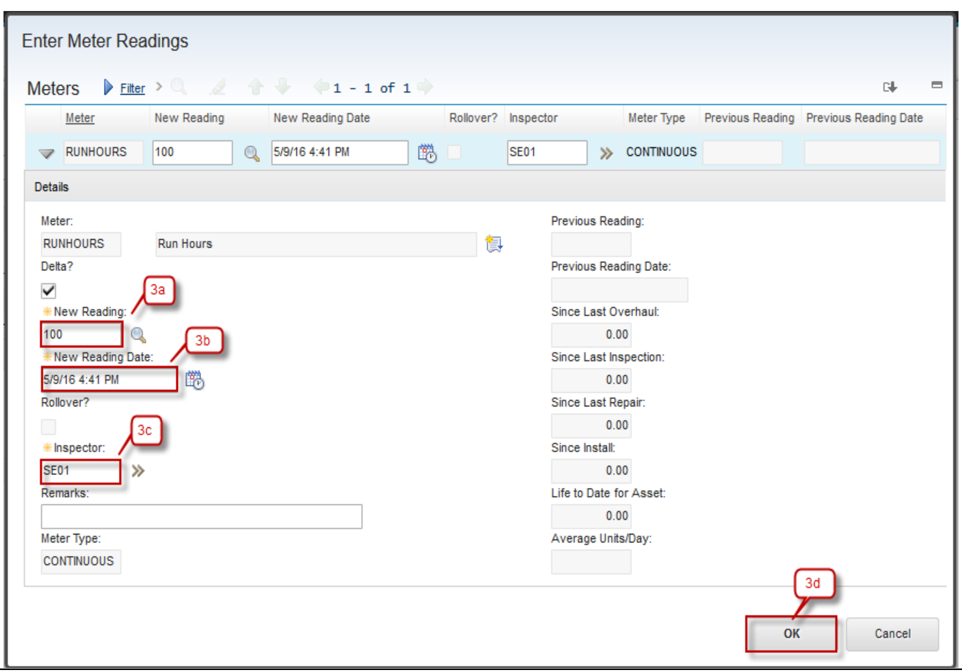


3a. Input value New Reading.

3b. New Reading Date automatically fill when input New Reading.

3c. Inspector automatically fill with user login.

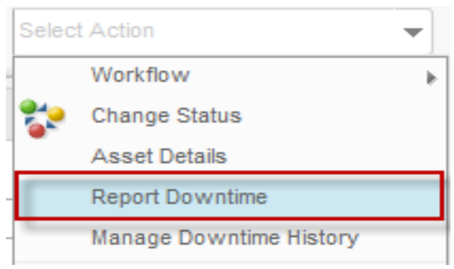
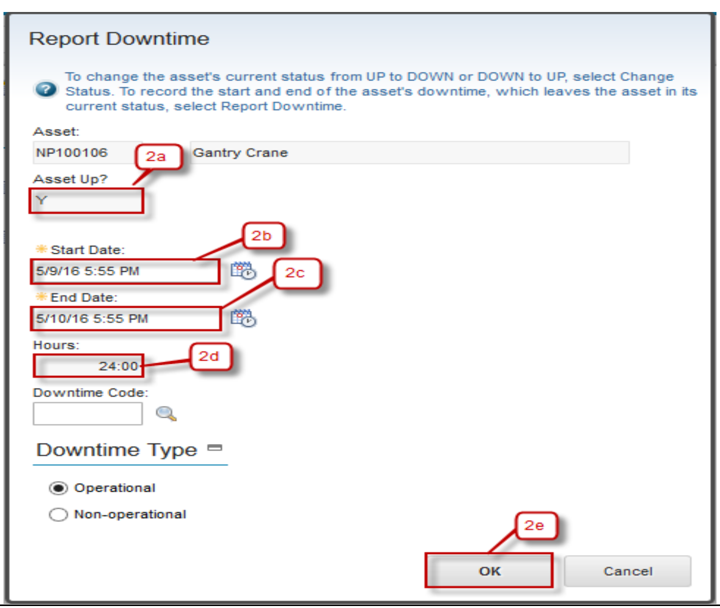
3d. Click **OK.**



2.5.5 Downtime

Downtime is another feature that available in an asset. This feature can be used to determine and track how long an asset has been out of service. With this feature, user can be informed about how effective an asset is. A field of total accumulated downtime that ever happens to the asset is available in **Asset** application too.

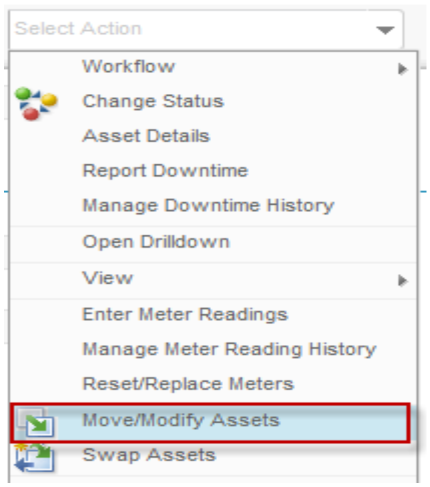
Downtime is reported in **Assets** application using **Report Downtime** action. In this action, user should define how long a downtime occurs to the asset, the start time, and the end time. As a note, this report downtime feature also available in **Work Order Tracking** application. Below are step by step to report downtime of an asset:

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1. Select one of Asset. Select Action → Report Downtime</td><td>
</td></tr>
<tr><td colspan="2">2a. Make sure default Asset Up? is Y.
2b. Start Date usually automatically fill with current date, but it’s editable.
2c. Input End Date until when the asset will be up again.
2d. Value of Hours is interval between Start Date and End Date.
2e. Click OK.

</td></tr>
</table>

2.5.6 Move Assets

When an asset need to be moved from a location to another location, for example, from operating location to repair location, user can do that from **Asset** application via **Move/Modify Assets** action. Besides locations, parent asset and condition code (for rotating asset) can be changed through this action too. Below are step by step to move location of an asset:

<table>
<tr><th>Step</th><th>Action</th></tr>
<tr><td>1. Select one of Asset. Select Action → Move/Modify Assets</td><td>



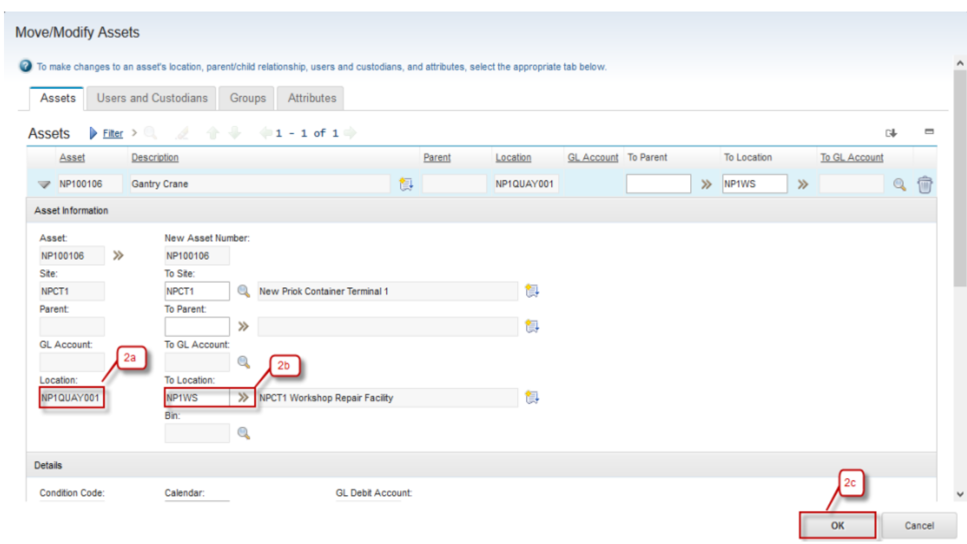
</td></tr>
<tr><td colspan="2">

2a. Current Location before being moved.

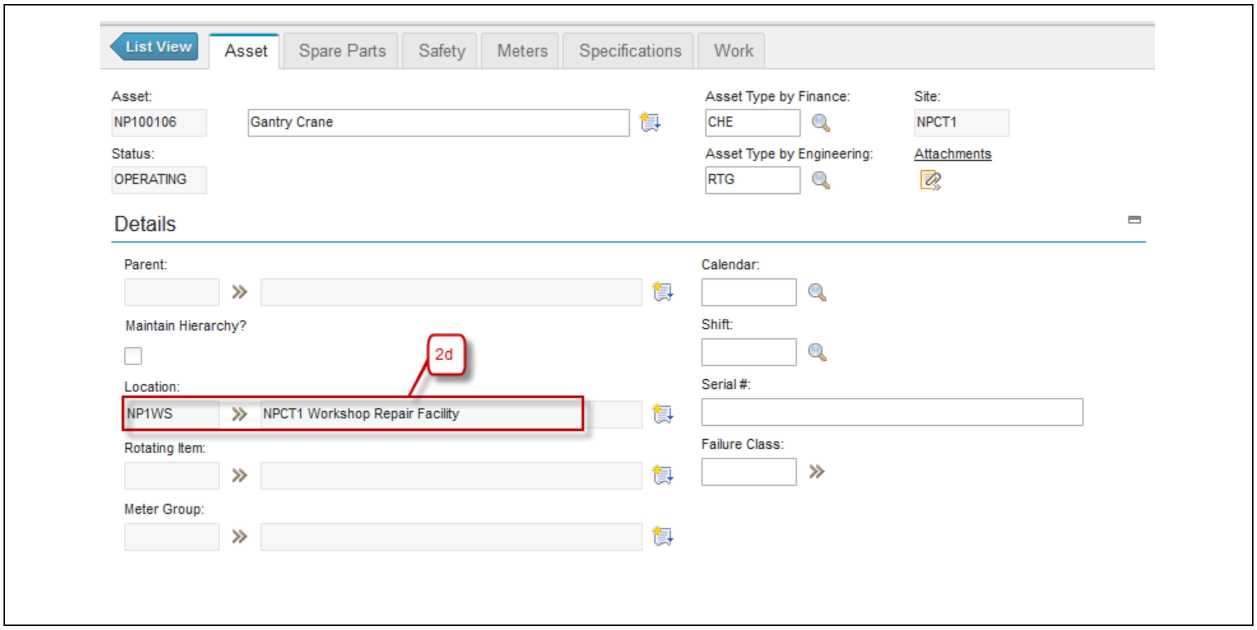
2b. Field To Location where the Asset move.

2c. Click OK.

2d. Asset already move to New Location.



</td></tr>
</table>



2.6 Exercise Phase